



Engineering Mathematics III

ENGE702

Lecture 14: Systems of Differential Equations

Systems of Differential Equations

Example: Two tanks each initially hold 100 litres of fluid. The fluid in tank 1 is pure water, whereas 30kg of salt has been dissolved in pure water in tank 2.

The solutions are exchanged between the tank at a rate of 5 litres per minute (assume the tanks are stirred so the solutions are uniform in both tanks).

Let $y_1(t)$ describe the amount of salt in tank 1 at time t and $y_2(t)$ describe the amount of salt in tank 2 at time t .

How long until there is half as much salt in tank 1 as there is in tank 2?

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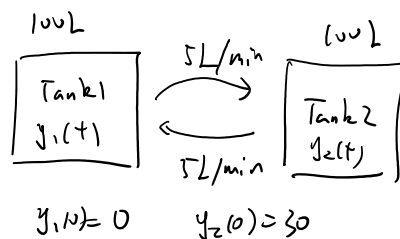
Let $y_1(t)$ describe the amount of salt in tank 1 at time t and $y_2(t)$ describe the amount of salt in tank 2 at time t .

How long until there is half as much salt in tank 1 as there is in tank 2?

First we must set up a model of the system. We have:

$$y_1' = \text{Inflow} - \text{Outflow} = \frac{5}{100}y_2 - \frac{5}{100}y_1$$

$$y_2' = \text{Inflow} - \text{Outflow} = \frac{5}{100}y_1 - \frac{5}{100}y_2$$



$$\frac{dy_1}{dt} = \text{Inflow from tank 2} - \text{outflow to tank 2}$$

$$\frac{dy_1}{dt} = \frac{y_2}{100} \times 5 - \frac{y_1}{100} \times 5 = 0.05y_2 - 0.05y_1$$

$$\frac{dy_2}{dt} = \text{Inflow from tank 1} - \text{outflow to tank 1}$$

$$\frac{dy_2}{dt} = \frac{y_1}{100} \times 5 - \frac{y_2}{100} \times 5 = 0.05y_1 - 0.05y_2$$

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0.05y_2 - 0.05y_1 \\ 0.05y_1 - 0.05y_2 \end{pmatrix} = \begin{pmatrix} -0.05 & 0.05 \\ 0.05 & -0.05 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$y_1(0) = 0 \quad y_2(0) = 30$$

$\Rightarrow$ Find eigenvalues and eigenvectors of $\begin{pmatrix} -0.05 & 0.05 \\ 0.05 & -0.05 \end{pmatrix}$ to get solutions $y_1(t), y_2(t)$

Systems of Differential Equations

This gives a system of differential equations:

$$\begin{aligned} y_1' &= -0.05y_1 + 0.05y_2 \\ y_2' &= 0.05y_1 - 0.05y_2, \end{aligned}$$

where $y_1(0) = 0$, $y_2(0) = 30$

Exercise: Solve this initial value problem and answer the original question.

(hint: there will be half as much salt in tank 1 as there is in tank 2 when tank 1 contains $\frac{1}{3}$ the total amount of salt).

look for solution $\Rightarrow y_1(t)$ and $y_2(t)$

Systems of Differential Equations

We will now show how to convert a second order homogeneous differential equation with constant coefficients to a system of two first order differential equations with constant coefficients.

Suppose we have a differential equation

$$ay'' + by' + cy = 0.$$

The trick to converting it comes from defining the vector

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} y \\ y' \end{pmatrix}.$$

Then

$$\mathbf{v}' = \begin{pmatrix} v_1' \\ v_2' \end{pmatrix} = \begin{pmatrix} y' \\ y'' \end{pmatrix}.$$

Systems of Differential Equations

Without loss of generality, assume $a = 1$. Note that rearranging our second order equation we get

$$y'' = -cy - by'.$$

We can then set up the system

$$\mathbf{v}' = \begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} y' \\ -cy - by' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix} = A\mathbf{v}$$

In this case the top row of the matrix A is always 0 1, and the bottom row coefficients come from the original second order equation.

If we have initial conditions $y(0) = \alpha$ and $y'(0) = \beta$, we can apply these too, and get

$$\mathbf{v}(0) = \begin{pmatrix} y(0) \\ y'(0) \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Systems of Differential Equations

Example: Solve the initial value problem

$$y'' + 5y' + 4y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

by converting it to a system of two first order differential equations.

Systems of Differential Equations

Example: Solve the initial value problem

$$y'' + 5y' + 4y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

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Another application of system of ODEs is to transform second order ODE with constant coefficients into system of ODEs.

For 2nd ODE:

$$ay'' + by' + cy = 0$$

$$\Rightarrow y'' = -\frac{b}{a}y' - \frac{c}{a}y$$

Suppose the two unknown functions are

$$y_1(t) = y(t), \quad y_2(t) = y'(t).$$

$$\text{so } y_1'(t) = y_2(t)$$

$$y_2'(t) = -\frac{b}{a}y_2(t) - \frac{c}{a}y_1(t)$$

$$\Rightarrow \begin{pmatrix} y_1'(t) \\ y_2'(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$$

A system of ODEs

Example: $y'' + 5y' + 4y = 0$ to a system of ODEs.

Let $y_1(t) = y(t)$ and $y_2(t) = y'(t)$, so

$$y_1'(t) = y_2(t)$$

$$y_2'(t) = -5y_2 - 4y_1$$

$$\Rightarrow \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -4 & -5 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$IC: y(0) = 0, y'(0) = 1 \Rightarrow y_1(0) = 0, y_2(0) = 1.$$

Systems of Differential Equations

Example: Solve the initial value problem

$$y'' + 5y' + 4y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

by converting it to a system of two first order differential equations.

We set

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} y \\ y' \end{pmatrix},$$

and get

$$\mathbf{v}' = \begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} y' \\ -4y - 5y' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -4 & -5 \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix}$$

with initial condition

$$\mathbf{v}(0) = \begin{pmatrix} y(0) \\ y'(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Exercise: Solve this initial value problem.

$$I : y_1(0) = 0, \quad y_1'(0) = 1 \Rightarrow y_1(0) = 0, \quad y_2(0) = 1.$$

Exercises

- Find the general solution for the following system of equations:

$$\begin{aligned} y_1' &= y_1 - 3y_2 \\ y_2' &= 0y_1 - 2y_2 \end{aligned}$$

- Solve the initial value problem

$$y'' - 3y' + 2y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

by converting it to a system of two first order equations.

Text Book References

Chapters 4.0, 4.1